

Group Quiz 01

Instructions

- The group quiz requires submission of two components: a report and a program. The report should contain a detailed analysis.
- The report should be well-organized and clear, with appropriate use of figures, text, and terminology. Clarity and organization of the report will contribute 2 points to the total score.
- The program should be well-written, generic (if possible), and reusable for similar problems. The code must be well-commented and clear to ensure understanding and correctness.
- The report and program should be submitted through a Google Form, which will be linked shortly. The deadline for submission is 5:00 PM on March 20, 2025. Late submissions will not be accepted.
- Grading will be based on the depth of the analysis, the quality of the program (8 points) and the clarity and organization of the report (2 points).
- It is a group project, and all team members should contribute to the analysis and coding. An additional viva voce could be conducted (if deemed necessary) to assess if every member has contributed to the project.
- Any source used for making the analysis or program must be cited. Lack of proper citation will be considered plagiarism.

1. Compute the response of a series RL circuit to a square wave input as shown in Fig. 1, both numerically and analytically. For the numerical approach, use appropriate numerical techniques to solve the differential equation governing the RL circuit's behavior. For the analytical approach, utilize Fourier Series to derive the response of the circuit. The square wave is characterized by a duty ratio, denoted by the factor α . You are free to choose the values of resistance R and inductance L for your analysis.

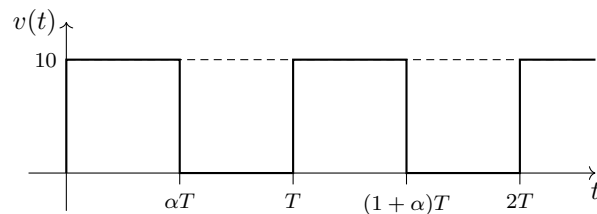


Fig. 1: Square wave with a duty ratio α

Analyze the circuit's behavior obtained from both numerical and analytical methods. In your analysis, apply the relevant concepts from the course related to circuits and differential equations to explore the response and draw comparisons between the two approaches.